



is carried out adiabatically in a PFR packed with a catalyst. Pure A enters the reactor at a volumetric flow rate of $20 \text{ dm}^3/\text{s}$, at a pressure of 10 atm, and a temperature of 450 K.

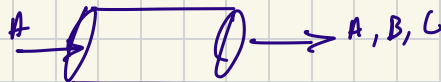
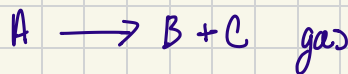
Additional information:

$$\begin{aligned} C_{PA} &= 40 \text{ J/mol} \cdot \text{K} & C_{PB} &= 25 \text{ J/mol} \cdot \text{K} & C_{PC} &= 15 \text{ J/mol} \cdot \text{K} \\ H_A^\circ &= -70 \text{ kJ/mol} & H_B^\circ &= -50 \text{ kJ/mol} & H_C^\circ &= -40 \text{ kJ/mol} \end{aligned}$$

All heats of formation are referenced to 273 K.

$$k = 0.133 \exp \left[\frac{E}{R} \left(\frac{1}{450} - \frac{1}{T} \right) \right] \frac{\text{dm}^3}{\text{kg-cat} \cdot \text{s}} \quad \text{with } E = 31.4 \text{ kJ/mol}$$

- Plot and then analyze the conversion and temperature down the plug-flow reactor until an 80% conversion (if possible) is reached. (The maximum catalyst weight that can be packed into the PFR is 50 kg.) Assume that $\Delta P = 0.0$.
- Vary the inlet temperature and describe what you find.
- Plot the heat that must be removed along the reactor ($\dot{Q}$ vs. V) to maintain isothermal operation.
- Now take the pressure drop into account in the PBR with $\rho_b = 1 \text{ kg/dm}^3$. The reactor can be packed with one of two particle sizes. Choose one.
 $\alpha = 0.019/\text{kg-cat}$ for particle diameter D_1
 $\alpha = 0.0075/\text{kg-cat}$ for particle diameter D_2
- Plot and then analyze the temperature, conversion, and pressure along the length of the reactor. Vary the parameters α and P_0 to learn the ranges of values in which they dramatically affect the conversion.
- Apply to this problem one or more of the six ideas discussed in Table P-4 in the Complete preface-introduction on the Web site (<http://www.umich.edu/~elements/6e/toc/Preface-Complete.pdf>).



$$\begin{aligned} \bullet v_0 &= 20 \text{ dm}^3/\text{s} & \bullet P &= 10 \text{ atm} \\ \bullet T &= 450 \text{ K} & \bullet C_{PA} &= 40 \text{ J/mol} \cdot \text{K} \\ \bullet C_{PB} &= 25 \text{ J/mol} \cdot \text{K} & \bullet C_{PC} &= 15 \text{ J/mol} \cdot \text{K} \\ \bullet H_A^\circ &= -70 \text{ kJ/mol} & \bullet H_B^\circ &= -50 \text{ kJ/mol} \\ \bullet H_C^\circ &= -40 \text{ kJ/mol} & \bullet T_{ref} &= 273 \text{ K} \end{aligned}$$

$$\begin{aligned} \bullet k(T) &= 0.133 \exp \left[\frac{E}{R} \left(\frac{1}{450} - \frac{1}{T} \right) \right] \frac{\text{dm}^3}{\text{kg} \cdot \text{cat} \cdot \text{s}} \\ \bullet E &= 31.4 \text{ kJ/mol} \end{aligned}$$

$$1) \quad r_A = -k(T) C_A \quad C_A = C_{A0} (1-X) \left(\frac{1}{1-\varepsilon X} \right) \left(\frac{T_0}{T} \right) \quad \bullet \varepsilon = 1$$

$$\frac{dX}{dW} = \frac{-r_A}{F_{A0}} = \frac{k(T) C_{A0} \left(\frac{1-X}{1+\varepsilon X} \right) \left(\frac{T_0}{T} \right)}{F_{A0}}$$

$$0 = \sum_{\alpha} F_{A0} h_{\alpha}(T_0) - \sum_{\alpha} F_{\alpha} h_{\alpha}(T)$$

$$0 = \underbrace{\sum_{\alpha} F_{A0} (h_{\alpha}(T_0) - h_{\alpha}(T))}_{C_{PA} (T_0 - T)} - \underbrace{\sum_{\alpha} v_{\alpha} h_{\alpha}(T) F_{A0} X}_{\Delta_r H}$$

$$\Delta_r H(T) = \Delta_r H(T_{ref}) + \cancel{\Delta_r C_p (T - T_0)} = -20 \text{ kJ/mol}$$

$$T = T_0 - \frac{\Delta_r H F_{A0} X}{\sum_{\alpha} F_{A0} C_{PA}}$$

$$\bullet \sum_{\alpha} F_{A0} = F_{A0}$$

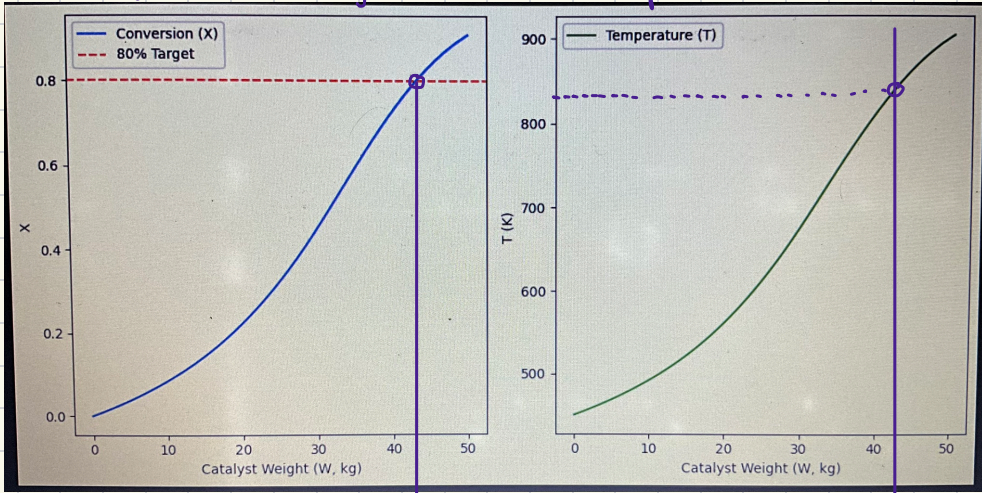
$$F_{A0} = \frac{PV}{RT} = \left(\frac{10 \text{ atm}}{460 \text{ K}} \right) \left(\frac{20 \text{ dm}^3}{\text{s}} \right) \left(\frac{\text{mol} \cdot \text{K}}{0.08206 \text{ dm}^3 \cdot \text{atm}} \right) = 5.42 \text{ mol/s}$$

• use $T = 850 \text{ K}$

$$\frac{dT}{dW} = \frac{dT}{dx} \left(\frac{dx}{dW} \right) = - \frac{\Delta_r H}{C_{PA}} \left(\frac{k(T) C_{A0} \left(\frac{1-x}{1+x} \right) \left(\frac{T_0}{T_i} \right)}{F_{A0}} \right)$$

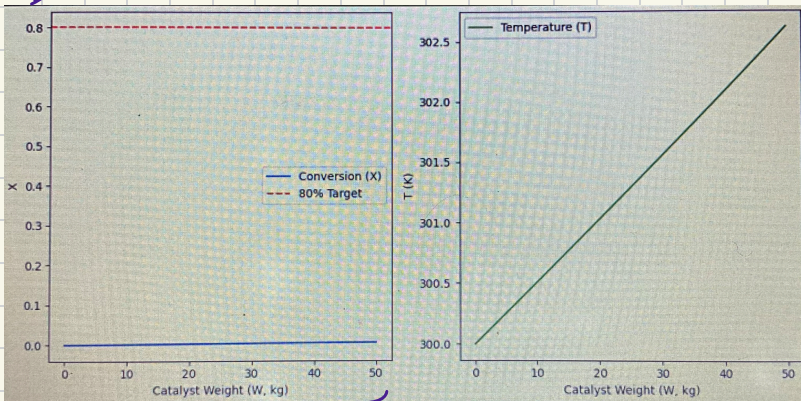
Conversion vs Catalyst wt

Temp vs Catalyst wt

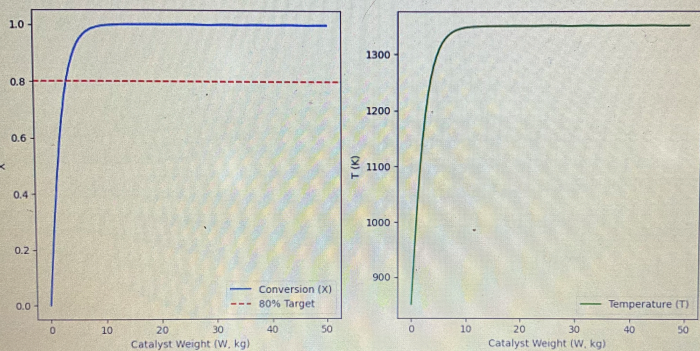


@ 80% conversion can be achieved with 43.2 kg of catalyst & will reach 850.5 K

2)



Conversion @ T_0 (800 K):
the rxn does not proceed
the temp profile is negligible & the temp rises due to the heat from the system



the conversion @ T_0 (850K): the Rxn reaches a conversion of 0.8 with less catalyst ($w = 3397$ kg-cat) & reaches higher temperatures like $T = 1250K$ at this conversion

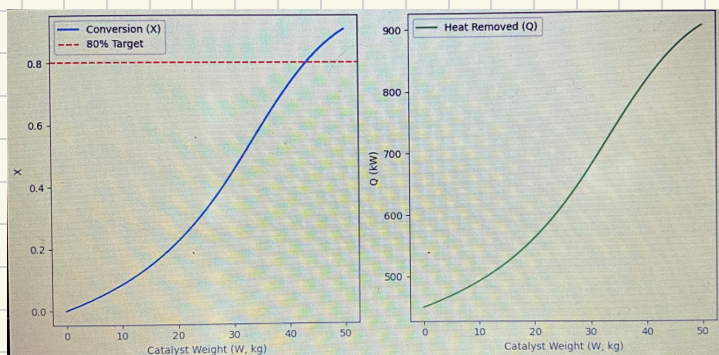
3) to keep the reactor isothermal we must remove the heat from the system since the rxn is exothermic ($\Delta H = -20 \text{ kJ/mol}$) we have

$$0 = \sum F_{in} C_{p,i} (T_0 - T) - \Delta_r H F_{A0} X \quad (\text{ignore } T_0 - T \text{ as rxn is iso thermal})$$

we must add $\dot{Q}$ if we removing heat making our eqn

$$0 = -\Delta_r H F_{A0} X + \dot{Q} \rightarrow \dot{Q} = \Delta_r H F_{A0} X$$

to plot $\dot{Q}$ vs V multiply $\frac{dX}{dW} = -\frac{r_A}{F_{A0}}$ by $\Delta_r H F_{A0}$ & set $T = T_0$ & $\frac{dT}{dW} \rightarrow \frac{d\dot{Q}}{dW} = -\Delta_r H F_{A0} \frac{dX}{dW}$



this is our plots

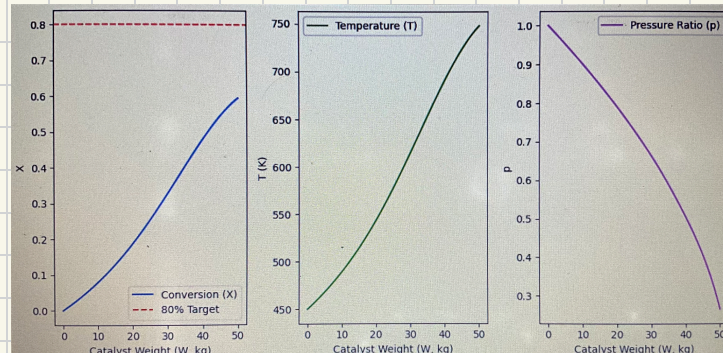
4) Accounting for the press drop in PBR with $\rho_0 = 1 \text{ kg/dm}^3$ & choosing $\alpha = 0.01 \text{ 1/kg-cat}$ for the particle diameter D_p

$$\frac{dP}{dW} = -\frac{\alpha}{2P} (1 - \epsilon X) \left(\frac{T_0}{T} \right) \quad \& \text{concentration changes}$$

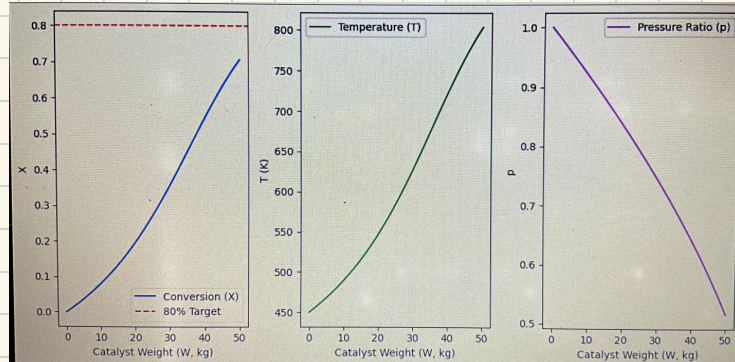
to $C_A = C_{A0} (1 - X) \left(\frac{1 - \epsilon X}{1 - \epsilon} \right) \left(\frac{T_0}{T} \right)^p$

5)

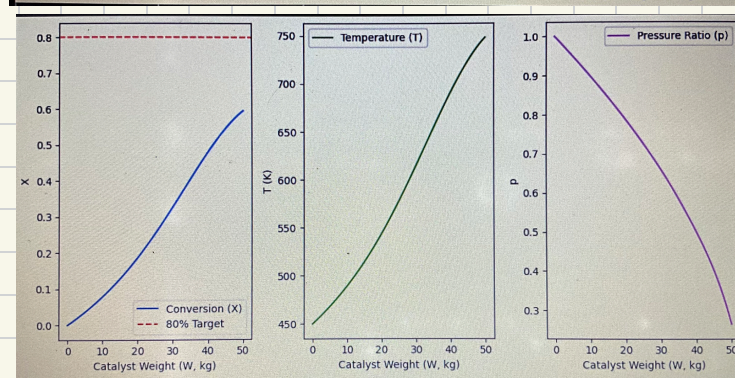
graph 1) is for the base case
X, T, & P vs W



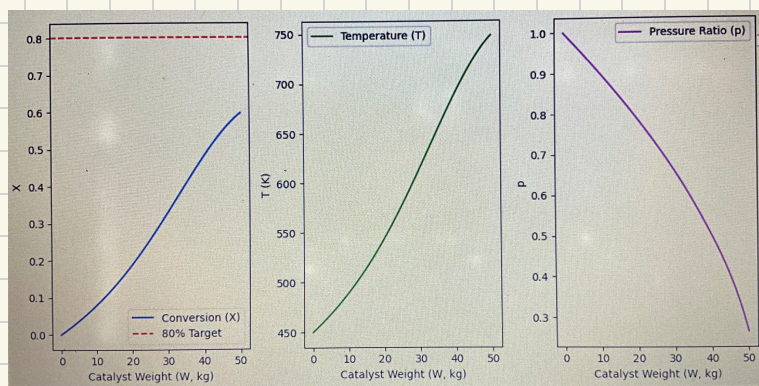
graph 2) when $\alpha = 0.015 \text{ 1/kg-cat}$
{ cant go above $\alpha = 0.0195$ }
X & T increase *
P vs W become linear

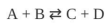


graph 3) when $P_0 = 5 \text{ atm}$
the conversion $\uparrow$ slightly



graph 4) when $P_0 = W_0 \text{ atm}$
the conversion
doesn't really change
Varying α causes the
most changes where
 P_0 only causes slight
changes





- adiabatic
- $T_{\min} = 27^\circ\text{C}$
- feed = equimolar A & B
- $x_{\text{eq}} = 0.9999$
- series of PFR with interstage cooling

is carried out adiabatically in a series of staged packed-bed reactors with interstage cooling (see Figure 11-5). The lowest temperature to which the reactant stream may be cooled is 27°C . The feed is equal molar in A and B, and the catalyst weight in each reactor is sufficient to achieve 99.9% of the equilibrium conversion. The feed enters at 27°C and the reaction is carried out adiabatically. If four reactors and three coolers are available, what conversion may be achieved?

Additional information:

$$\Delta H_{R,27}^\circ = 30000 \text{ cal/mol} \quad A C_{pA} = C_{pB} = C_{pC} = C_{pD} = 25 \text{ cal/mol} \cdot \text{K}$$

$$K_e(50^\circ\text{C}) = 500000 F_{A0} = 10 \text{ mol A/min}$$

First prepare a plot of equilibrium conversion as a function of temperature. (Partial ans: $T = 360 \text{ K}$, $X_e = 0.984$; $T = 520 \text{ K}$, $X_e = 0.09$; $T = 540 \text{ K}$, $X_e = 0.057$)

$$T = T_0 + \frac{-\Delta H_{R,27}^\circ X}{\sum F C_p} = T_0 \frac{-\Delta H_{R,27}^\circ X}{C_{pA} + C_{pB}} = 300 + 1600X$$

$$K_e = 500000 \exp \left[\frac{-30000}{1787} \left(\frac{1}{323} - \frac{1}{T} \right) \right]$$

T	X
300	1
360	0.984
520	0.09
540	0.057

